Python Engineering - Class 08 Theory Guide

Data Processing

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Boolean selection

A mask turns an engineering question into a vectorized logical expression.

Relationship / equation

```
ascending = v > 0
```

Connection to code

Instead of iterating row by row, the same condition is evaluated across an entire column.

2. Compound conditions

Engineering regions often need more than one condition.

Relationship / equation

```
(h > 50) & (v > 0)
```

Connection to code

Parentheses are important because NumPy uses element-wise logical operators rather than Python's scalar and/or.

3. Indices vs values

Filtering data and locating data are different operations.

Relationship / equation

```
x[mask] -> values; np.where(mask)[0] -> indices
```

Connection to code

Choose depending on whether the next calculation needs the measurements or their positions.

4. Finite differences

np.diff turns samples into sample-to-sample increments.

Relationship / equation

```
dt_k=t_{k+1}-t_k
```

Connection to code

The result has N-1 elements because N samples contain N-1 intervals.

5. Data augmentation

Derived quantities can be appended as additional columns to form a richer dataset.

Relationship / equation

```
[h,v,t] -> [h,v,t,energy]
```

Connection to code

This is the beginning of building engineering data pipelines.

Manual validation method

For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

Masks and data processing prepare filtering, statistics, sensitivity studies and post-processing.

Final summary

```
MODEL / CONCEPT
      |
      v
ALGORITHM
      |
      v
PYTHON IMPLEMENTATION
      |
      v
REFERENCE CHECK
      |
      v
ENGINEERING INTERPRETATION
```

> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.